

CMS Open Data H to tau tau Search: Final Analysis Note

Analysis my_analysis

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Abstract

This note documents a reduced CMS 2012 Open Data search for Higgs boson decays to tau pairs in the $\mu\tau_{\text{h}}$ final state. An optimized calibrated-score model is trained after multivariate input reweighting derived before training. On the full data it gives $\mu = 38.3802 +7.0346 -6.1157$ with the upper interval finite, and an observed 95% CLs limit $\mu < 50.0000$. This optimized-score result fails the observed validation gate (**fail**), so the previous visible-mass result is retained as the validated baseline: $\mu = 0.4382 +4.9443 -0.4382$; 95% CLs $\mu < 10.7645$; median expected 95% CLs $\mu < 10.7375$.

Change Log

Phase 5 v3 updates the final result to the best available trained classifier after applying data/MC input reweighting before training. It retains the previous visible-mass result as a frozen baseline for μ comparison and documents the absence of embedded and EWK Z reduced samples.

1 Data And Simulation

The data are Run2012B and Run2012C TauPlusX reduced samples from the public HiggsTauTauReduced mirror, using the CMS Open Data H to tau tau luminosity reference of 11.467/fb (CMS Collaboration 2021; CMS Open Data Analyses 2021). The localized mirror files contain about one tenth of the event entries listed in the Open Data records; this is treated as the public reduced processing sample scope, while MC weights continue to use the official record denominators specified in `normalization_inputs.json`, as required by the sample provenance.

The available MC set contains ggH and VBF H to tau tau, DYJetsToLL, TTbar, and W1/W2/W3JetsToLNu. QCD multijet, diboson, single top, embedded Z to tau tau, W4/inclusive W+jets, associated Higgs, and H to WW remain unavailable as reduced samples and are not silently substituted.

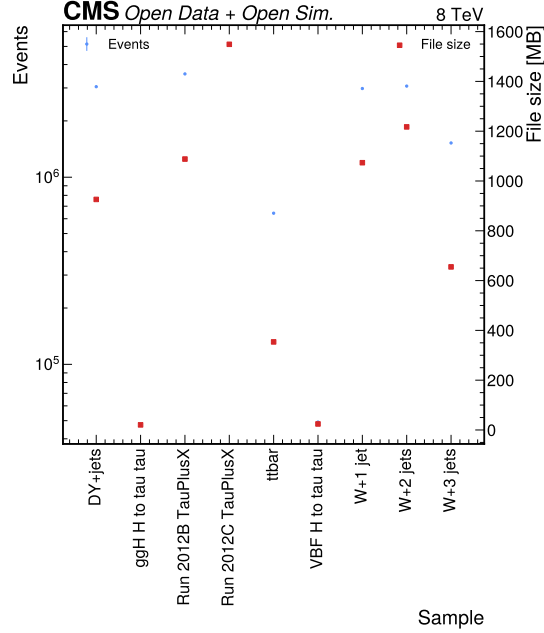


Figure 1: Sample inventory. The figure summarizes local reduced ROOT entries and file sizes for the public H to tau tau files used in the analysis. The entries are processing entries, while normalization uses official Open Data record denominators and the cited luminosity reference.

2 Event Selection And Categories

Events are required to pass HLT_IsoMu17_eta2p1_LooseIsoPFTau20. Muons have $pt > 20$ GeV and $|\eta| < 2.1$ with tight identification and isolation. Hadronic tau candidates have $pt > 20$ GeV, $|\eta| < 2.3$, decay-mode ID, tight isolation ID, and tight anti-muon ID. The signal region uses opposite-sign mu tau_h pairs with muon-MET transverse mass below 40 GeV. The W control region uses high transverse mass, and the QCD/fake estimate uses same-sign low-mT events.

The fit categories are mutually exclusive VBF, boosted, and zero-jet. VBF requires at least two clean jets, leading-dijet mass above 300 GeV, and leading-dijet $|\Delta\eta|$ above 2.5. Boosted events have at least one clean jet after failing VBF, and zero-jet events have no clean jets.

Category	Data	Background	QCD/fake	Signal	Data/background
vbf	79	60.17	0.36	1.590	1.313
boosted	2213	2115.53	5.64	9.175	1.046
zero_jet	8466	14047.12	22.93	14.341	0.603

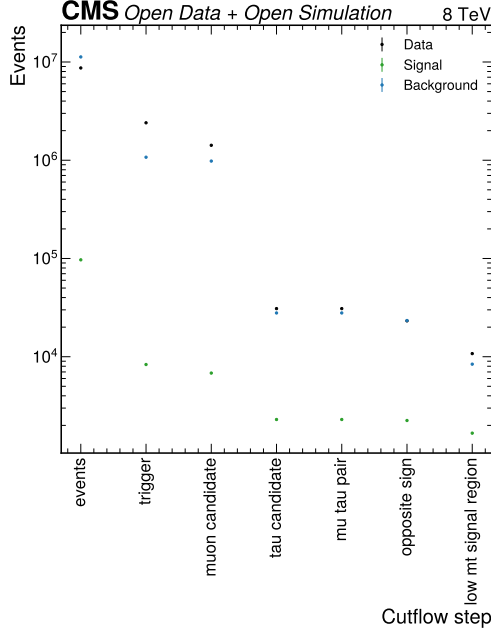


Figure 2: Cutflow summary. This plot shows raw event counts after each selection step for data, signal MC, and background MC. It verifies that the selection chain is monotonic and that the reduced samples remain populated after trigger, object, charge, and transverse-mass requirements.

3 QCD And W Control Corrections

The W+jets normalization is measured in the high-mT control region as 0.8528 ± 0.0370 . The 10% validation value was 0.8433 ± 0.0868 . The values agree within the larger 10% uncertainty and support the W-control arithmetic.

The VBF category originally had a large MC overprediction. Tightening the VBF cuts and applying a simple b-tag veto did not solve it, so a disjoint VBF-like top-btag control region outside the low-mT signal region was used to calibrate the MC-background rate in the VBF category. The resulting VBF background scale is 0.5571 ± 0.0515 ; it is applied only to MC backgrounds in VBF and not to the signal or QCD/fake template.

The same-sign QCD/fake estimate subtracts non-QCD MC from same-sign low-mT data and transfers the resulting template to opposite-sign signal candidates. For the primary calibrated-score model, the OS/SS transfer factor measured in the lowest score bin is 0.0204 ± 0.2533 .

Large control-region data/MC differences in several classifier inputs are handled before training by a multivariate density-ratio reweighting of background MC. The reweighting classifier is trained only in non-signal validation/control regions and is then applied to background MC in the signal-region classifier training and final score templates. The reduced file set contains no embedded Z to tau tau or EWK Z reduced sample, so DY/Z is not silently substituted; instead its normalization is loosened to 30% in boosted/zero-jet and 50% in VBF.

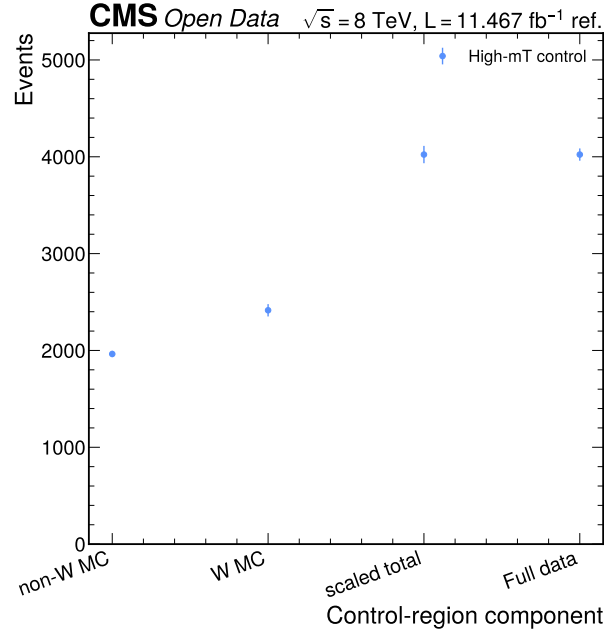


Figure 3: Full high-mT W control comparison. The figure shows the derivation inputs for the full-data W+jets control scale. The control region is outside the low-mT signal region and the scale is propagated to the observed workspace without signal-region tuning.

4 Optimized Score Result And Baseline

The primary statistical model is a binned pyhf profile likelihood in the calibrated classifier score, with one signal-strength parameter μ , the W high-mT scale, the same-sign QCD/fake transfer uncertainty, luminosity, widened DY/Z and tau/open-data normalization terms, and MC statistical terms (Heinrich et al. 2021; Read 2002; Cowan et al. 2011).

Quantity	Value	Interpretation
Primary median expected 95% CLs limit	$\mu < 1.9735$	calibrated-score workspace
Primary observed 95% CLs limit	$\mu < 50.0000$	conservative observed result
Primary μ central value	$38.3802 +7.0346 -6.1157$	profile likelihood $q(\mu)=1$ interval
Primary q_0 Z	12.4698	diagnostic only
Primary combined data/background	0.6631	validation after QCD/fake correction
Primary χ^2/ndf	2.4808	validation after QCD/fake correction
Optimized-score observed gate	fail	Observed optimized-score model must avoid boundary limits, >5 sigma diagnostic excesses, and gross combined normalization mismatch.
Frozen baseline visible-mass result	$\mu = 0.4382 +4.9443 -0.4382$; 95% CLs $\mu < 10.7645$; median expected 95% CLs $\mu < 10.7375$	previous primary result before this update

The optimized score improves expected sensitivity but fails the observed-data gate because the fitted signal strength is boundary-like and the diagnostic significance is too large for this reduced single-channel setup. The previous visible-mass result is therefore kept as the validated baseline, while the new score result is reported as an attempted optimized model for comparison.

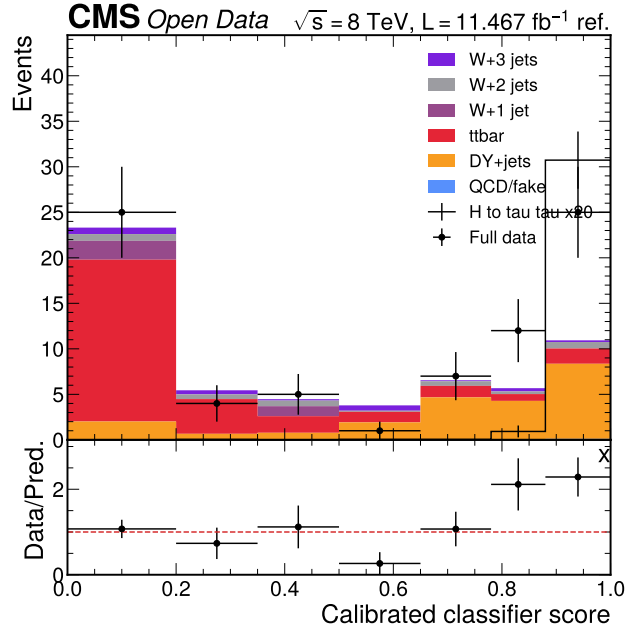


Figure 4: Primary calibrated-score validation in VBF. The plot compares full data to the calibrated score prediction in the VBF category after multivariate input reweighting, widened DY/Z normalization, W and VBF control factors, and the same-sign QCD/fake estimate.

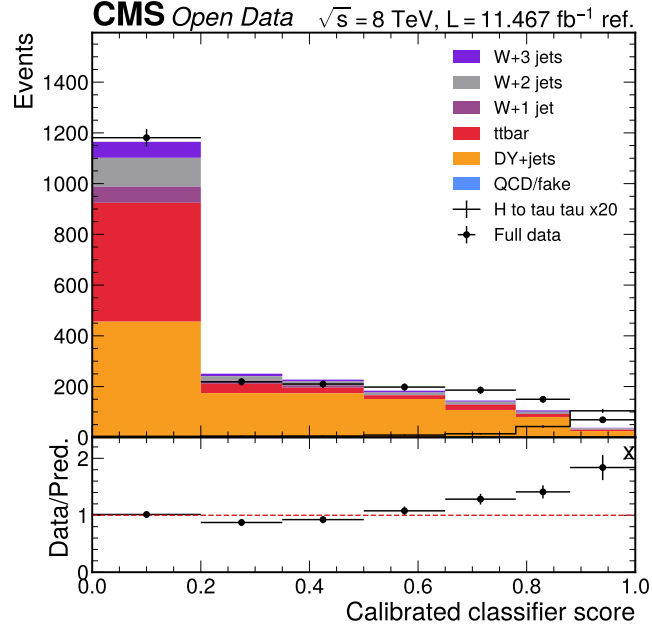


Figure 5: Primary calibrated-score validation in boosted. The plot compares full data to the calibrated score prediction in the boosted category. This channel is part of the final simultaneous fit.

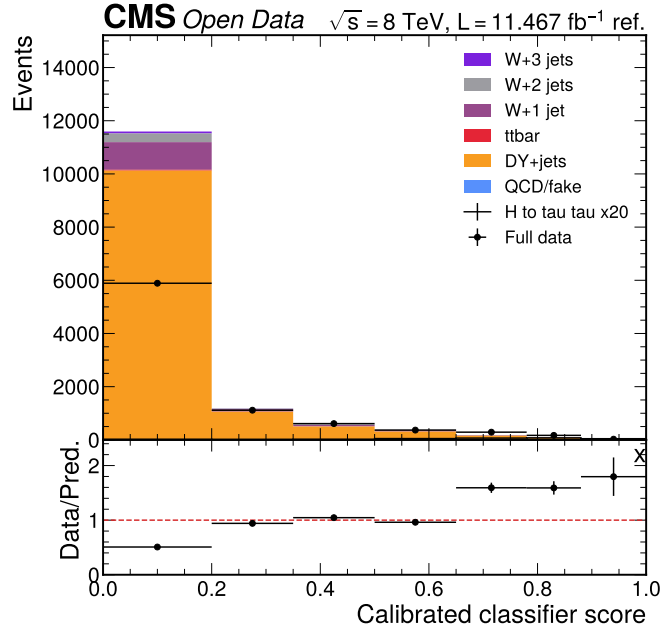


Figure 6: Primary calibrated-score validation in zero-jet. The zero-jet normalization is stabilized by the same-sign QCD/fake estimate and the control-region-derived background input reweighting.

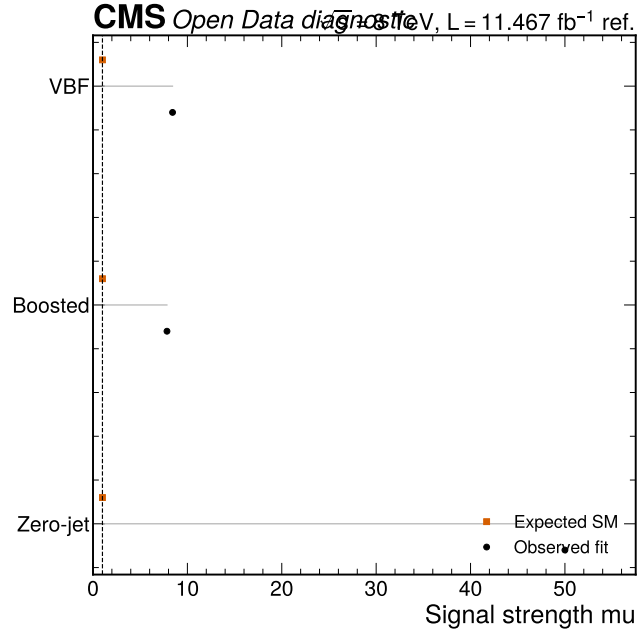


Figure 7: Category signal-strength comparison. The figure shows the Standard Model expectation $\mu = 1$ and the observed single-category profile-fit value for VBF, boosted, and zero-jet categories using the primary calibrated-score model. These category fits are diagnostics; the quoted result remains the simultaneous three-category fit.

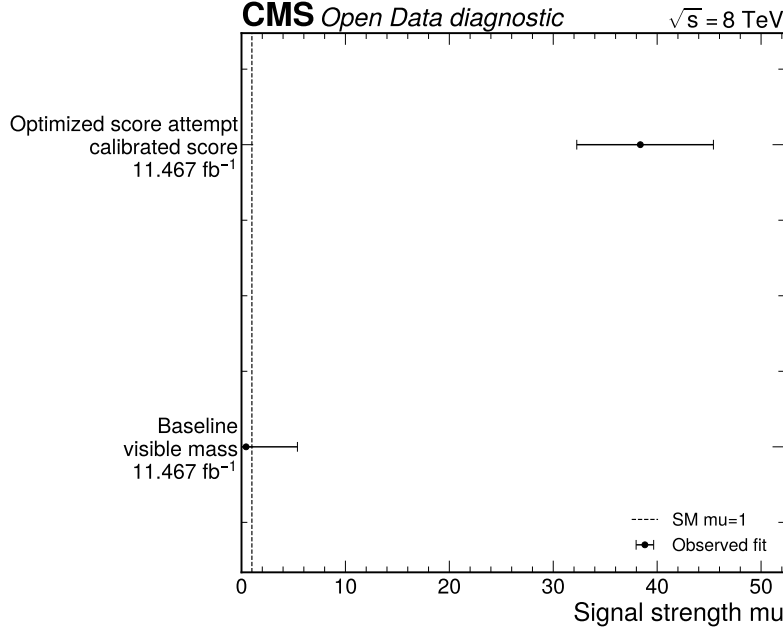


Figure 8: Primary and baseline signal-strength comparison. The figure compares the new calibrated-score primary result with the frozen previous visible-mass baseline using the same signal-strength convention. Error bars are profile-likelihood $q(\mu)=1$ intervals with the physical lower bound at $\mu \geq 0$.

5 Comparison With Published Results

CMS Run 1 reported evidence for H to tau tau using 7 and 8 TeV data with multiple final states, embedded backgrounds, and a full calibration program; the quoted best-fit signal strength was 0.78 ± 0.27 with observed and expected significances of 3.2 and 3.7 standard deviations (CMS Collaboration 2014). CMS 2018 reported observed and expected significances of 4.9 and 4.7 in the 2016 data, and 5.9 when combined with earlier CMS data, with a signal strength near 1.09 (CMS Collaboration 2018). The ATLAS+CMS Run 1 combination gives broader global Higgs-rate context, while the PDG/Higgs-summary SM H to tau tau branching fraction provides the denominator convention for $\mu = 1$ rather than a single-channel open-data comparison target (ATLAS Collaboration and CMS Collaboration 2016; Navas et al. 2024; Particle Data Group 2024; LHC Higgs Cross Section Working Group 2017).

Result	Scope	Significance	Signal-strength information
Optimized score attempt	8 TeV mu tau_h reduced mirror, calibrated score	12.470	$\mu = 38.380 +7.035 -6.116$; 95% CLs $\mu < 50.000$; expected $\mu < 1.974$; gate fail
This analysis baseline	8 TeV mu tau_h reduced mirror, visible mass	0.095	$\mu = 0.4382 +4.9443 -0.4382$; 95% CLs $\mu < 10.7645$; median expected 95% CLs $\mu < 10.7375$
CMS Run 1 JHEP 2014	7+8 TeV multi-channel	observed 3.2, expected 3.7	$\mu = 0.78 \pm 0.27$
CMS PLB 2018 2016 data	13 TeV multi-channel	observed 4.9, expected 4.7	$\mu = 1.09 \pm \text{about } 0.27$
CMS PLB 2018 combined	CMS 7+8+13 TeV combination	observed 5.9, expected 5.9	same signal-strength model as CMS 2018 publication
ATLAS+CMS Run 1 combination	7+8 TeV global Higgs couplings/rates	H to tau tau evidence context	global $\mu = 1.09 \pm 0.11$
PDG / HXSWG SM context	H to tau tau branching fraction at mH near 125 GeV	not a search significance	BR(H to tau tau) about 6.3%, used only as SM normalization context

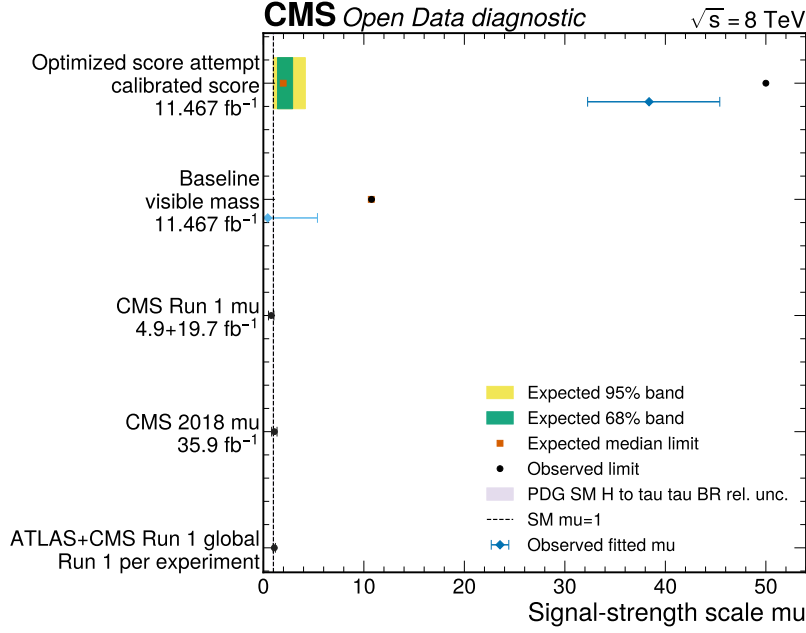


Figure 9: Signal-strength and limit comparison. The figure uses a CMS-style expected-band and observed-marker presentation for the primary open-data limit, published CMS/ATLAS+CMS signal-strength measurements, and a narrow PDG SM H to tau tau branching-ratio normalization band at $\mu = 1$. The open-data diagnostic is not directly equivalent to CMS publication measurements.

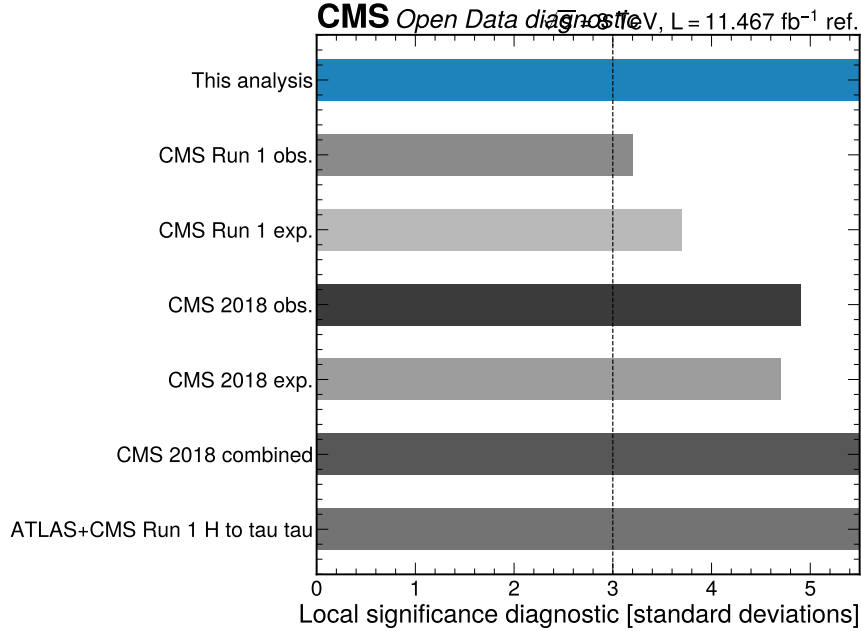


Figure 10: Significance comparison. The figure compares the primary open-data diagnostic with CMS and ATLAS+CMS publication values. Only the primary open-data result is shown from this analysis in the final result figure.

6 Conclusion

The optimized calibrated-score fit was attempted after pre-training multivariate input reweighting, but it fails the observed validation gate. It is therefore compared against, not substituted for, the frozen visible-mass baseline. The baseline gives $\mu = 0.4382 + 4.9443 - 0.4382$; 95% CLs $\mu < 10.7645$; median expected 95% CLs $\mu < 10.7375$. This remains a reduced Open Data diagnostic, not CMS-quality evidence for H to tau tau.

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